

Week 8 Summer Homework: FFT, Sampling, and Smart-Grid Signals

Goal: connect physical phenomena, time-series measurements, frequency spectra, and MATLAB implementation.

AI policy for this homework. You may use AI or web search to learn syntax, understand MATLAB documentation, debug error messages, and compare plotting styles. You may **not** ask AI to write the required MATLAB code for you. Your submitted scripts must be your own manual coding work. If you use AI for explanation, write one sentence describing what it helped you understand.

Mathematical reminder. A repeating physical motion can often be modeled as

$$x(t) = A \sin(2\pi f t + \phi) + c,$$

where A is amplitude, f is frequency in Hz, ϕ is phase, and c is an offset. A sensor samples this continuous motion at sample rate F_s to create a sequence $x[0], x[1], \dots, x[N-1]$. The FFT converts this time-domain sequence into frequency-domain information. To avoid aliasing, the practical sampling rule is

$$F_s > 2f_{\max}.$$

Monday: Physical Signal and Time-Domain Plot

Choose one physical phenomenon that can be represented by a signal, such as spring motion, sound, vibration, AC voltage, solar generation, or household electricity load. Create a MATLAB script that simulates at least two seconds of data.

1. Define a sample rate `Fs`, time vector `t`, amplitude, frequency, and phase.
2. Generate one clean sine-wave signal using `sin`.
3. Plot the time-domain signal with clear axis labels, title, line width, and grid.
4. In three or four sentences, explain what the signal represents physically. What does the amplitude mean? What does the frequency mean?

Hints, not code: search the official MATLAB documentation for creating vectors with colon notation, `sin`, `plot`, `xlabel`, `ylabel`, `title`, and `grid`.

Tuesday: FFT Spectrum and Dominant Frequency

Build a mixed signal with two rhythms and a small amount of random noise. For example, combine a slow wave, a faster wave, and noise. Then use `fft` to reveal the hidden frequency ingredients.

1. Submit the time-domain plot of your mixed signal.
2. Submit the one-sided FFT amplitude spectrum with the horizontal axis in Hz, not just array index.
3. Clearly identify the two largest nonzero-frequency peaks in your spectrum.
4. Explain in three sentences why the FFT spectrum can be easier to interpret than the raw time-domain plot.

Important: Your spectrum should use the sample rate to build a real frequency axis. A plot whose x -axis is only “index” is incomplete.

Wednesday: Manual Dominant-Frequency Function

Write a MATLAB function named `find_dominant_frequency(x, Fs)`. The function should return the strongest nonzero frequency in the signal.

1. Compute `X = fft(x)`.
2. Convert FFT bins into frequencies in Hz.
3. Build the one-sided amplitude spectrum.
4. Use a `for` loop to find the largest nonzero-frequency amplitude.
5. Return the frequency of that largest peak.
6. Test your function on a signal whose true frequency you already know, such as 5 Hz or 35 Hz.

Manual coding rule. Do not use `findpeaks`. Do not use `max` to directly produce the final answer. You may use `fft`, `abs`, `length`, `floor`, array indexing, and a manually written `for` loop.

Thursday: Sampling Theorem and Aliasing Experiment

Use MATLAB to simulate the same physical signal under two different sample rates: one good sample rate and one bad sample rate.

1. Choose a true signal frequency `f_true`.
2. Choose `Fs_good > 2*f_true` and `Fs_bad < 2*f_true`.
3. Plot the well-sampled signal and the badly sampled points on the same figure.
4. Write four or five sentences explaining how the bad sample rate can create a false story about the original physical motion.

Friday: Feynman Technique Post

Make a post explaining FFT to a beginner using one physical example.

Submission checklist: Monday physical-signal script and plot; Tuesday mixed-signal plot and FFT spectrum; Wednesday `find_dominant_frequency` function and test; Thursday aliasing experiment; Friday Feynman note.